

SIH 2026 EWS-NER: Full System Functions & Architecture Guide

AI-Powered Landslide Early Warning, Remote Sensing Telemetry & Real-Time Disaster Intelligence

1. System Architecture & Core Technology Stack

The EWS-NER platform is an end-to-end multi-tier disaster early warning architecture designed for mountainous and landslide-vulnerable terrain in the North Eastern Region (NER) and Western Ghats. It synthesizes live satellite radar topography (NASA SRTM 30m DEM), numerical weather prediction (Open-Meteo & OpenWeatherMap), sub-surface soil moisture telemetry, and computer vision hazard verification.

Layer / Module	Technology Stack	Core Functionality
Web Frontend	React 18, TypeScript, Tailwind CSS, Vite	Citizen safety portal & officer dashboards
3D Simulation	Three.js, WebGL Shader Engine	Interactive 3D mountain elevation & debris runoff path
Computer Vision	HTML5 Canvas, Edge-Tensor Scanner	Crowdsourced crack, mudflow & fracture detection
Voice Assistant	Web Speech API (TTS & Speech-to-Text)	Vernacular voice alarms & hands-free reporting
AI Susceptibility	Python 3.11, FastAPI, XGBoost	Multi-factor mathematical vulnerability index (0.0 - 1.0)
Detour Routing	NetworkX Graph Engine, Dijkstra	Dynamic road rerouting: NH-766 blocked -> SH-59 safe detour
Relief Shelter	Real-time Resource Allocator	Live bed occupancy, ration days, and SDRF tracking
Offline SOS	Bluetooth Low Energy (BLE) Mesh	Zero-internet emergency distress multi-hop relay
Database	PostgreSQL 16 with PostGIS	Spatial geometry, hazard polygons, historical records

2. AI Mathematical Models & Decision Formulas

2.1. XGBoost Multi-Factor Landslide Susceptibility Index (S)

Function: `compute_xgboost_susceptibility(slope, rain_24h, rain_72h, soil_moisture)`

Calculates the continuous vulnerability probability by weighting normalized terrain slope, antecedent rainfall, and soil moisture saturation:

Equation: $S = 0.35 * (\text{Slope} / 50^\circ) + 0.30 * (R_{24} / 200\text{mm}) + 0.20 * (\Theta_{\text{soil}} / 0.60 \text{ m}^3/\text{m}^3) + 0.15 * (R_{72} / 350\text{mm})$

- **RED (Critical):** $S \geq 0.70$ or $R_{24} \geq 110\text{mm}$ → Automatic emergency siren, highway closure, immediate evacuation.
- **AMBER (Elevated):** $0.40 \leq S < 0.70$ → Pre-warning advisory, relief shelter readiness, heavy transit restrictions.
- **GREEN (Normal):** $S < 0.40$ → Stable terrain conditions, continuous monitoring.

2.2. NetworkX Dynamic Evacuation Routing Algorithm

Function: `compute_evacuation_routing(region_name, lat, lon, risk_score)`

Models the regional transit network as a graph $G(V, E)$. When $S \geq 0.65$, primary hazard links (e.g. NH-766) are dynamically severed, and the shortest safe evacuation corridor (SH-59 Relief Bypass) is computed via Dijkstra pathfinding.

2.3. Multi-Hazard 5-Factor AI Prioritization Agent

Function: `runAIPriorityAgent(alerts)`

Ranks concurrent landslide emergencies across districts (Wayanad, Munnar, Guwahati, Shillong, Aizawl) to optimize NDRF rescue team dispatch:

Formula: $P = 0.35 * RiskScore + 0.25 * RainNorm + 0.20 * PopDensity + 0.15 * FieldReports + 0.05 * Recency$

3. Frontend Components & Interactive Features

Component / Hook	File Path	Detailed Functional Description
CitizenPortal.tsx	frontend/src/pages/CitizenPortal.tsx	Primary citizen dashboard displaying real-time risk status, 24h/72h rainfall telemetry
AiVisionScanner.tsx	components/report/AiVisionScanner.tsx	Computer Vision photo hazard scanner. Draws AI bounding boxes on tension crack
Terrain3DVisualizer.tsx	components/map/Terrain3DVisualizer.tsx	Three.js WebGL 3D terrain viewer. Shows mountain elevation mesh, slope heat gra
ShelterResourcePanel.tsx	components/panels/ShelterResourcePanel.tsx	Relief shelter tracker showing real-time bed capacity, occupancy percentage, food
OfflineSosMesh.tsx	components/panels/OfflineSosMesh.tsx	Zero-Internet BLE Mesh SOS broadcaster. Generates distress packets (coordinate
useVoiceAssistant.ts	hooks/useVoiceAssistant.ts	Multilingual Speech Synthesis & Speech-to-Text hook. Reads emergency warnings
useAlertSound.ts	hooks/useAlertSound.ts	Web Audio API synthesizer generating real 400Hz-900Hz sweeping emergency sir
usePermissions.ts	hooks/usePermissions.ts	Manages browser push notification & GPS permissions with non-blocking 4-second
GisMapDashboard.tsx	components/map/GisMapDashboard.tsx	Interactive Leaflet GIS map with zero-error OpenStreetMap tiles, red hazard polyg
ReportFormPage.tsx	frontend/src/pages/ReportFormPage.tsx	Crowdsourced incident submission with AI Vision photo verification, category grid,

4. Live Remote Sensing & Satellite Telemetry Ingestion

Data Source	API Provider	Parameters & Metrics Ingested
Ground Weather Station	OpenWeatherMap Live API	Station Name, Live Temperature (21.3°C), Humidity (91%), Wind Velocity
Satellite Topography	NASA SRTM 30m Global DEM	Point digital elevation (876.5 meters) and 3D slope gradient calculation
Sub-surface Soil Moisture	Open-Meteo European Agency	0-1cm topsoil layer saturation (0.267 m³/m³) and hourly precipitation
Disaster Alert Standard	NDMA SACHET / CAP 1.2	Common Alerting Protocol (CAP) compliant emergency broadcast schema
GIS Base Layers	OpenStreetMap & CARTO	High-performance vector tiles with 0 authentication errors

5. Live Cloud Deployment & Verified Endpoints

- Live Citizen Safety Web Application: <https://frontend-eta-rouge-44.vercel.app>
- AI Computer Vision & Voice Reporting: <https://frontend-eta-rouge-44.vercel.app/report>
- FastAPI AI & Telemetry Microservice: <https://ews-ai-engine.onrender.com>
- Spring Boot REST Gateway: <https://ews-backend-gateway.onrender.com>
- Managed PostgreSQL with PostGIS: ews-postgres-db (Singapore Cluster)
- Native Android Application: EWS-Landslide-AI-Early-Warning.apk (4.5 MB)